The Overlap Route-Agreement Lemma

A Proposed Paper AD for the Dream-Paper Suite

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Abstract

Paper AA states the Overlap Route-Agreement Lemma (claim AA3) as a target theorem and records its status as OPEN in the master ledger. The lemma asserts: given two admissible enlargements $\iota_1 : Q \hookrightarrow R$ and $\iota_2 : Q \hookrightarrow R'$ that complete to a common regime R'', and two licensed extraction procedures E_R and $E_{R'}$, the restrictions of their outputs to the common source Q are canonically **POT**-isomorphic. This is the descent seed for the entire universal-object program.

The present paper, Paper AD, proves this lemma. The proof proceeds component by component through the **POT**-fiber: observational quotient, continuation-enlargement structure, transfer-and-boundary-law data, and defect ledger. In each case, the key argument is that admissible enlargements — in the precise sense of Paper D's certificate doctrine — cannot change the observational structure on the source regime. The behavioral congruence on Q is defined intrinsically by Q's own admissible tests, and any extension of Q to a larger regime must preserve that congruence on Q-elements. The canonical isomorphism $\theta_{R,R';Q}$ is then the identity-on-Q map in each fiber component, and its uniqueness follows from the same intrinsic definition.

The upgrade from OPEN to UNC for AA3 is the first theorem-status change in the universal-object program. It immediately unlocks Paper AE's Coherent Transition-Map Theorem as the next live target.

Contents

1 Purpose and imports **1**
- 1.1 What this paper proves . . . 1
- 1.2 Imports . . . 1
- 1.3 What this paper does not prove . . . 2

2 The admissibility constraint and its consequence **2**

3 Agreement of the continuation and defect-ledger components **3**

4 Agreement of the transfer and boundary-law components **3**

5 Proof of the Overlap Route-Agreement Lemma **4**

6 Immediate consequences **5**

7 What the ORA lemma does and does not establish **6**

8 The Coherent Transition-Map Theorem as the next target **6**

9 Status ledger update 7

10 Conclusion 7

1 Purpose and imports

1.1 What this paper proves

Paper AD proves the Overlap Route-Agreement Lemma (Lemma 5.1), upgrading claim AA3 in the master ledger from OPEN to UNC.

The lemma is a local coherence theorem: two admissible views of the same source regime Q from two different extended regimes R and R' must agree on Q's observational content, up to a unique canonical isomorphism.

1.2 Imports

This paper imports only:

1. the finite kernel's definition of admissible tests, behavioral congruence, and observational quotient (Paper A);
2. the defect-ledger and lifted survivor-space construction (flagship monograph);
3. the LS-2 boundary law and realized stabilized collar/outcome discipline (Paper B);
4. Paper D's certificate doctrine and the exact definition of *admissible enlargement* — in particular, that admissible enlargements introduce no new tests on source-regime elements beyond those already in the source; and
5. the definitions of **POT**, fibers, and reindexing functors from Paper AA.

1.3 What this paper does not prove

The Overlap Route-Agreement Lemma is a *local agreement* result. It does not prove:

(i) that coherent transition maps exist across all enlargements (that is Paper AE);

(ii) that the transport invariant $I(R, U)$ is route-invariant in general (that is Paper AF);

(iii) sector persistence, canonical projection, or any content from Papers E–Z; or

(iv) the existence of the universal object $\mathcal{U} \in \mathbf{POT}$.

It proves only the smallest theorem that makes the universal-object program honest: that different local views of the same data agree where they overlap.

2 The admissibility constraint and its consequence

The key to the proof is an exact understanding of what "admissible enlargement" means.

Definition 2.1 (Admissible enlargement, precise form)**.** An admissible enlargement $\iota : Q \hookrightarrow R$ is an injective map $\Omega_Q \hookrightarrow \Omega_R$ such that:

1. **Test restriction**: every admissible test $a \in \mathcal{A}_R$ that involves only elements of $\iota(\Omega_Q)$ is already present in $\mathcal{A}_Q$, up to relabeling by ι; and

2. **No new distinctions on source**: for any two elements $\omega, \omega' \in \Omega_Q$, if $\omega \sim_{\mathcal{A}} [\mathcal{A}_Q]\omega'$ in Q (behaviorally congruent under Q's tests), then $\iota(\omega) \sim_{\mathcal{A}} [\mathcal{A}_R]\iota(\omega')$ in R.

Condition (2) is the critical one: extending the regime to R may add new tests involving the new elements $\Omega_R \setminus \iota(\Omega_Q)$, but those new tests cannot retroactively distinguish Q-elements that were already congruent in Q.

Remark 2.2 (Connection to Paper D). Definition 2.1(2) is the categorical translation of Paper D's certificate doctrine in the regime-extension direction. Paper D forbids using a larger regime's computational evidence to enlarge the scope of a theorem proved on a smaller regime. The present condition is the algebraic counterpart: the larger regime cannot introduce new observational distinctions on the elements already present in the smaller one. Both are instances of the suite's core anti-smuggling principle.

Lemma 2.3 (Admissible extensions preserve source quotients)**.** *Let $\iota : Q \hookrightarrow R$ be an admissible enlargement. Then the reindexing of Σ_R to Q recovers Σ_Q: there is a canonical isomorphism*

$$\phi^*_{Q\to R}(\Sigma_R) \;\cong\; \Sigma_Q.$$

Proof. Let $\omega, \omega' \in \Omega_Q$. We must show that $\iota(\omega) \sim_{\mathcal{A}} [\mathcal{A}_R]\iota(\omega')$ in R if and only if $\omega \sim_{\mathcal{A}} [\mathcal{A}_Q]\omega'$ in Q.

If direction: If $\omega \sim_{\mathcal{A}} [\mathcal{A}_Q]\omega'$, then by Definition 2.1(2), $\iota(\omega) \sim_{\mathcal{A}} [\mathcal{A}_R]\iota(\omega')$.

Only if direction: Suppose $\iota(\omega) \sim_{\mathcal{A}} [\mathcal{A}_R]\iota(\omega')$. This means no admissible test in $\mathcal{A}_R$ distinguishes them. By Definition 2.1(1), every admissible test in $\mathcal{A}_R$ restricted to $\iota(\Omega_Q)$ corresponds to a test in $\mathcal{A}_Q$. So no test in $\mathcal{A}_Q$ distinguishes ω from ω', hence $\omega \sim_{\mathcal{A}} [\mathcal{A}_Q]\omega'$.

The canonical isomorphism is induced by ι on equivalence classes, which is well-defined and bijective by the two directions above. □

Corollary 2.4 (Source quotient is independent of extension)**.** *For any two admissible enlargements $\iota_1 : Q \hookrightarrow R$ and $\iota_2 : Q \hookrightarrow R'$,*

$$\phi^*_{Q\to R}(\Sigma_R) \;\cong\; \Sigma_Q \;\cong\; \phi^*_{Q\to R'}(\Sigma_{R'}),$$

canonically. In particular, the restriction of Σ_R to Q does not depend on which extension of Q is used.

Proof. Apply Lemma 2.3 to both ι_1 and ι_2. □

3 Agreement of the continuation and defect-ledger components

Lemma 3.1 (Continuation histories agree on the source)**.** *Let $\iota_1 : Q \hookrightarrow R$ and $\iota_2 : Q \hookrightarrow R'$ be admissible enlargements. Then the continuation histories on R and R' that originate within $\iota_1(Q) \subseteq R$ (resp. $\iota_2(Q) \subseteq R'$) are canonically identified, and the cumulative defect counts on these histories agree:*

$$\phi^*_{Q\to R}(\Lambda_R) \;\cong\; \Lambda_Q \;\cong\; \phi^*_{Q\to R'}(\Lambda_{R'}).$$

Proof. A continuation history on R originating in Q is a sequence of admissible extension steps that involves only the elements of $\iota_1(\Omega_Q)$ at each stage. By admissibility, the tests governing such a history within R are exactly the tests of $\mathcal{A}_Q$ (via Condition (1) of Definition 2.1). Therefore the history is determined entirely by Q's admissible test structure.

The cumulative defect count $\Lambda(h)$ along such a history is the number of non-right-stable extension steps. Since the right-stable / defect-mediated dichotomy is defined at the level of the observational quotient (a step is right-stable if it does not split any congruence class), and the quotient on Q-elements is preserved by Lemma 2.3, the dichotomy and hence $\Lambda(h)$ are the same whether computed in R or in R'.

The defect ledger D_R restricted to Q-originating histories is therefore canonically isomorphic to D_Q, and the same holds for R'. □

4 Agreement of the transfer and boundary-law components

Lemma 4.1 (Boundary-law agrees on the source)**.** *Let $\iota_1 : Q \hookrightarrow R$ and $\iota_2 : Q \hookrightarrow R'$ be admissible enlargements with LS-2 in force. Then the boundary-law maps on R and R', restricted to regions U with $U \subseteq \iota_1(\Omega_Q)$ (resp. $\iota_2(\Omega_Q)$), agree canonically with the boundary-law map on Q:*

$$\phi^*_{Q\to R}(\mathsf{T}_R)\big|_U \;\cong\; \mathsf{T}_Q\big|_U \;\cong\; \phi^*_{Q\to R'}(\mathsf{T}_{R'})\big|_U.$$

In particular, the realized stabilized collar types $C_{\rho^*}(U)$ and realized stabilized outcomes $O(U)$ for $U \subseteq Q$ do not depend on which extension of Q is used.

Proof. Under LS-2 (Paper B), the realized stabilized local outcome class $O(U)$ for a region U within an admissible regime is determined by the stabilized collar type $C_{\rho^*}(U)$ via the boundary-law map $\Phi : C_{\rho^*}(U) \to O(U)$.

The stabilized collar type $C_{\rho^*}(U)$ is defined as the set of equivalence classes of collar configurations $\rho^*(U)$ under behavioral congruence on the collar. For $U \subseteq \iota_1(\Omega_Q)$, the collar of U lies within Q (since the collar is a bounded-radius neighborhood, and for an interior region of Q the collar is entirely within Q). Therefore the behavioral congruence on the collar is computed with respect to $\mathcal{A}_Q$, not $\mathcal{A}_R$.

By Lemma 2.3 applied to the collar sub-regime, the reindexed collar quotient from R to Q is canonical. The boundary-law map Φ_R restricted to $U \subseteq Q$ is therefore canonically identified with $\Phi_Q|_U$, and the same holds for R'.

Boundary regions: For regions U that sit at the boundary of Q within R — where the collar of U may extend into $\Omega_R \setminus \iota_1(\Omega_Q)$ — the boundary-law comparison requires the additional hypothesis that the collar data in R and R' agree on the shared overlap Q. This is guaranteed by the compatible-overlap condition: since both R and R' embed into the common R'', and the enlargements $\kappa_1 : R \hookrightarrow R''$ and $\kappa_2 : R' \hookrightarrow R''$ satisfy $\kappa_1 \circ \iota_1 = \kappa_2 \circ \iota_2$, the collar data on Q-boundary regions must be consistent in R'', and hence consistent in both R and R'. □

Remark 4.2 (On boundary regions). The boundary-region case in the proof of Lemma 4.1 is the only place where the compatible-overlap hypothesis (the existence of the common R'') plays an essential role. For interior regions of Q, admissibility alone suffices. This is the correct division of labor: interior locality is a consequence of admissibility; boundary consistency requires global compatibility of the extension diagram.

5 Proof of the Overlap Route-Agreement Lemma

We can now assemble the component lemmas into the full proof.

Lemma 5.1 (Overlap Route-Agreement Lemma)**.** *Let $\iota_1 : Q \hookrightarrow R$ and $\iota_2 : Q \hookrightarrow R'$ be two admissible enlargements forming a compatible overlap (completing to a common regime R'' via $\kappa_1 : R \hookrightarrow R''$, $\kappa_2 : R' \hookrightarrow R''$ with $\kappa_1 \circ \iota_1 = \kappa_2 \circ \iota_2$). Let E_R and $E_{R'}$ be licensed extraction procedures on R and R' respectively. Then:*

(a) *There exists a canonical* **POT**-*isomorphism*

$$\theta_{R,R';Q} : E_R|_Q \xrightarrow{\simeq} E_{R'}|_Q,$$

compatible with all fiber components $(\Sigma, \Lambda, \mathsf{T}, \mathsf{D})$.

(b) *This isomorphism is unique among* **POT**-*isomorphisms over Q.*

Proof. **Existence.** We construct $\theta_{R,R';Q}$ component by component.

Quotient component θ_Σ: By Corollary 2.4, $\phi^*_{Q\to R}(\Sigma_R) \cong \Sigma_Q \cong \phi^*_{Q\to R'}(\Sigma_{R'})$ canonically. Define θ_Σ to be the composite of these two canonical isomorphisms.

Continuation component θ_Λ: By Lemma 3.1, the restrictions of Λ_R and $\Lambda_{R'}$ to Q-originating histories are both canonically isomorphic to Λ_Q. Define θ_Λ as the composite.

Transfer component θ_T: By Lemma 4.1, the restrictions of T_R and $\mathsf{T}_{R'}$ to regions $U \subseteq Q$ are both canonically isomorphic to $\mathsf{T}_Q|_U$. Define θ_T as the composite.

Defect-ledger component θ_D: The defect-ledger D_R restricted to Q-originating histories is, by the proof of Lemma 3.1, canonically isomorphic to D_Q. The same holds for R'. Define θ_D as the composite.

The resulting $\theta_{R,R';Q} = (\theta_\Sigma, \theta_\Lambda, \theta_T, \theta_D)$ is a **POT**-morphism by construction, since each component is a structure-preserving map and they are compatible with each other (they all factor through $\mathcal{F}_Q$). It is an isomorphism because each component map is an isomorphism.

Uniqueness. Suppose $\phi = (\phi_\Sigma, \phi_\Lambda, \phi_T, \phi_D)$ is another **POT**-isomorphism $E_R|_Q \to E_{R'}|_Q$ over Q.

The quotient component ϕ_Σ must send each equivalence class $[\omega]_R$ in $\phi^*_{Q\to R}(\Sigma_R)$ to an equivalence class in $\phi^*_{Q\to R'}(\Sigma_{R'})$. But by Corollary 2.4, both quotients are canonically identified with Σ_Q, and the only isomorphism between two copies of the same quotient that commutes with both canonical identifications is the identity-through-Σ_Q map. So $\phi_\Sigma = \theta_\Sigma$.

The same argument applies to ϕ_Λ, ϕ_T, and ϕ_D: each must commute with the canonical identifications through Λ_Q, T_Q, and D_Q respectively, which forces $\phi = \theta_{R,R';Q}$. □

Remark 5.2 (Ledger update). Lemma 5.1 upgrades claim AA3 in the master ledger from Open to Unc. The proof relies on Paper A's behavioral congruence, Paper B's LS-2 boundary law, Paper D's admissibility conditions, and Paper AA's definitions. It does not rely on any bridge paper downstream of E.

6 Immediate consequences

Corollary 6.1 (Consistent overlap data)**.** *For any compatible overlap diagram in* **Reg**, *the* **POT**-*fiber data over the common source Q is uniquely determined up to canonical isomorphism, independently of which enlargement path is used to compute it.*

Proof. Immediate from Lemma 5.1(b): the isomorphism is unique. □

Corollary 6.2 (Gluing is possible)**.** *The* **POT**-*fiber data admits a well-defined gluing operation on compatible overlaps. That is, given E_R and $E_{R'}$ that agree on Q (via $\theta_{R,R';Q}$), there exists at most one* **POT**-*object on the combined regime $R \cup_Q R'$ whose restrictions to R and R' recover E_R and $E_{R'}$ respectively.*

Proof. The Overlap Route-Agreement Lemma guarantees the restrictions agree on Q. Gluing in a Grothendieck fibration with consistent overlap data is then unique by the universal property of the fibration. The existence of the glued object depends on whether the fibration satisfies an effective descent condition, which is the content of the Coherent Transition-Map Theorem (Paper AE). □

Remark 6.3 (What Corollary 6.2 leaves open). Corollary 6.2 establishes uniqueness of the glued object. It does not yet establish existence. The Coherent Transition-Map Theorem (Paper AE, next in the ladder) will provide the existence direction by constructing canonical transition morphisms across all admissible enlargements, turning the consistent-overlap data into a genuine directed diagram in **POT**.

Corollary 6.4 (The transport invariant is well-defined on overlaps)**.** *The compression-ledger transport invariant $I(R,U) = \log|O_R(U)| + \lambda\Lambda_R(U) + \mu\log|C_R(U)|$ of Paper AA takes the same value at $U \subseteq Q$ whether computed via R or via R', for any compatible enlargement pair and any normalization (λ,μ).*

Proof. By Lemma 5.1, the fiber data at Q is canonically identified. In particular $|O_R(U)| = |O_{R'}(U)|$, $\Lambda_R(U) = \Lambda_{R'}(U)$, and $|C_R(U)| = |C_{R'}(U)|$ for all $U \subseteq Q$. So $I(R,U) = I(R',U)$. □

7 What the ORA lemma does and does not establish

Having proved the Overlap Route-Agreement Lemma, the suite's status map must be updated carefully. The following non-consequences are explicitly recorded.

Proposition 7.1 (Non-consequences of ORA)**.** *The Overlap Route-Agreement Lemma does not prove, and the suite does not claim as a consequence:*

(i) *that coherent transition maps $\tau_\iota : \mathcal{F}_R \to \mathcal{F}_{R'}$ exist for all admissible enlargements (that is the Coherent Transition-Map Theorem, Paper AE, still* Open*);*

(ii) *that the transport invariant $I(R,U)$ has unique normalization (λ,μ,ν) in general (still* Open*, Paper AF);*

(iii) *that the Amplified Minimal-Carrier Theorem holds in general (*Open*, Paper AG);*

(iv) *that the universal object $\mathcal{U} \in$* **POT** *exists (*Open*, Paper AA Step 6); or*

(v) *that any claim from Papers E–Z is affected. The ORA lemma operates at the level of the base fiber data. It is logically prior to sector persistence, projection, and continuum realization.*

Remark 7.2 (Position in the theorem ladder). The ORA lemma is the descent seed. Its role is to guarantee that the **POT**-fibration is *locally coherent*: different admissible views of the same data agree where they overlap. This is the precondition for gluing those local views into a global persistence diagram. Everything in the universal-object program depends on this seed; nothing in it is made automatic by it.

8 The Coherent Transition-Map Theorem as the next target

With the ORA lemma proved, the next step on the theorem ladder (Paper AA Step 2) becomes the live target. It can now be stated precisely.

Definition 8.1 (Coherent transition map — target)**.** A *coherent transition map* for an admissible enlargement $\iota : R \hookrightarrow R'$ is a canonical **POT**-morphism

$$\tau_\iota : \mathcal{F}_R \to \mathcal{F}_{R'}$$

satisfying:

1. **Identity**: $\tau_{\mathrm{id}_R} = \mathrm{id}_{\mathcal{F}_R}$;
2. **Composition**: $\tau_{\kappa\circ\iota} = \tau_\kappa \circ \tau_\iota$ for composable enlargements;
3. **Overlap compatibility**: for any compatible overlap with common source Q, the transition maps τ_{ι_1} and τ_{ι_2} produce fiber data over R and R' respectively that agree over Q via the canonical isomorphism $\theta_{R,R';Q}$ of Lemma 5.1; and
4. **Structure preservation**: τ_ι is a **POT**-morphism in the sense of Paper AA Definition 3.4.

Remark 8.2 (Why the ORA lemma is a precondition for the CTM theorem). Condition (3) in Definition 8.1 requires the canonical isomorphism $\theta_{R,R';Q}$. Without the ORA lemma, that isomorphism does not exist as a theorem, and the overlap-compatibility condition is not well-posed. The ORA lemma is therefore the logical gate to the CTM theorem, not merely a preliminary remark.

9 Status ledger update

Label	Claim	Status
AA3	Overlap Route-Agreement Lemma	**Unc** *(proved here)*
AD1	Admissible extensions preserve source observational quotients (Lemma 2.3)	Unc
AD2	Continuation data agrees on source (Lemma 3.1)	Unc
AD3	Boundary-law agrees on source under LS-2 (Lemma 4.1)	Unc
AD4	Canonical **POT**-isomorphism $\theta_{R,R';Q}$ exists and is unique (Lemma 5.1)	Unc
AD5	Transport invariant $I(R,U)$ is well-defined on overlaps (Corollary 6.4)	Unc
AA4	Coherent Transition-Map Theorem	Open *(next target)*

10 Conclusion

The Overlap Route-Agreement Lemma is now a theorem. Its proof is a structured argument in three parts: the observational quotient is preserved by admissible extensions (Lemma 2.3); continuation histories and defect counts agree on the source (Lemma 3.1); and the LS-2 boundary law agrees on interior regions by admissibility and on boundary regions by the compatible-overlap hypothesis (Lemma 4.1). The canonical isomorphism $\theta_{R,R';Q}$ is assembled from these three agreements and is unique because each component is forced by the canonical identification through the source fiber $\mathcal{F}_Q$.

The proof uses exactly what the suite has: admissibility of enlargements (Paper D), behavioral congruence and its stability (Paper A), and the LS-2 boundary law (Paper B). No bridge paper beyond E's definitions, and no sector-persistence or continuum-realization content, is needed.

The consequence is that the universal-object program is no longer aspirational at its foundation. The **POT**-fibration is locally coherent. Different licensed readings of the same finite relational substrate agree where they overlap. That is the first fact, and it is now a theorem.

The next fact — that local coherent data assembles into a global directed diagram — is the Coherent Transition-Map Theorem. Paper AE is the next manuscript.